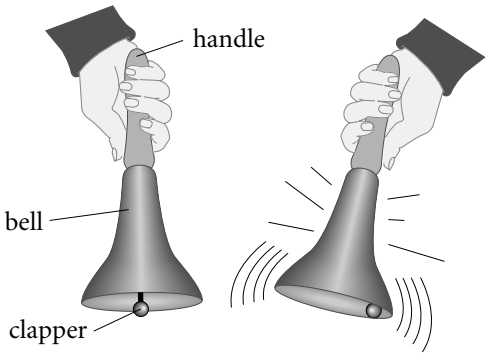
How are different sounds made?

1 Look at the picture of a handbell. When you shake it by the handle, the bell rings.



a What happens to the clapper when you shake the bell?

b What happens to the metal of the bell when it is struck?

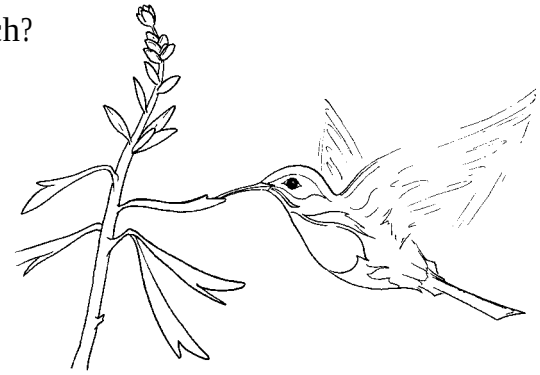
If the ringer puts a finger on the metal bell while shaking it, the bell makes a dull noise instead of a ring.

c Why doesn't the bell ring?

The bell is part of a set. Smaller bells have a higher pitch and larger bells have a lower pitch.

d Why will the size of the bell affect its pitch?

2 A humming-bird flaps its wings very fast. This makes a humming sound, which is how the bird gets its name.



What will happen to the pitch of the hum if the humming-bird:

a flaps its wings more slowly?

b flaps its wings more quickly?

3 Here are some CRO traces (graphs) of different sounds, from tuning forks and from musical instruments.

a Which of these sounds is the loudest?

b Which is the quietest?

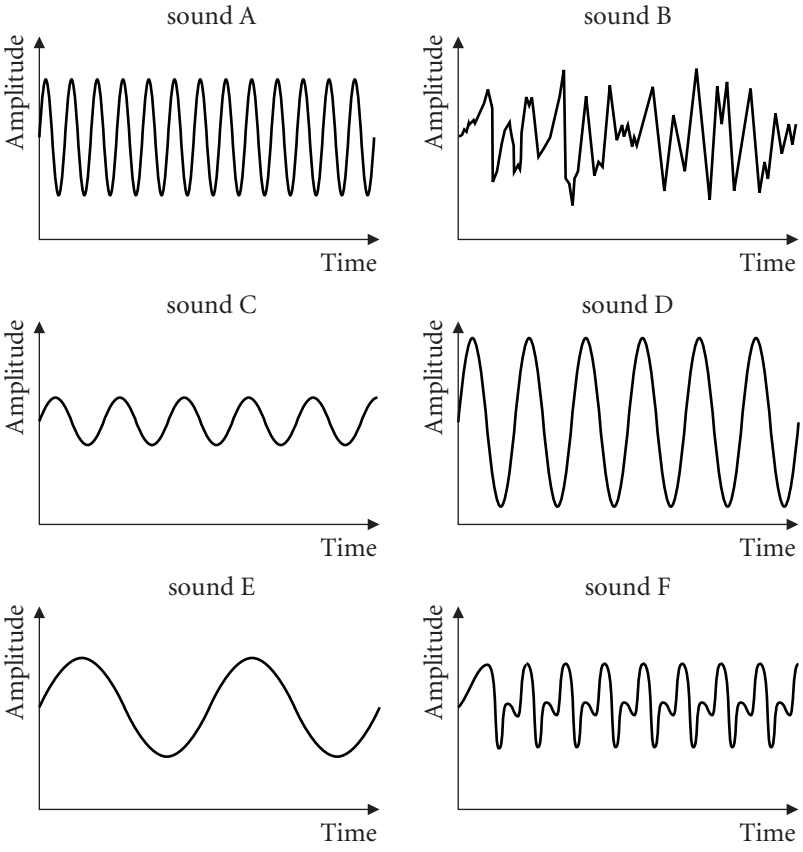
c Which of these sounds is the highest pitched?

d Which is the lowest pitched?

e Which sounds were made by tuning forks?

f Which sound has the largest amplitude?

g Which sound has the lowest frequency?



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How does sound travel through solids, liquids and gases?

1 Sometimes people put their ear to the door or wall to hear what is happening on the other side. American Indians put their ear to the ground to listen for horses. Explain how these actions help us to hear better.

2 There are lots of boats and ships with engines travelling across the oceans of the world. How might this affect communications between animals like dolphins and whales?

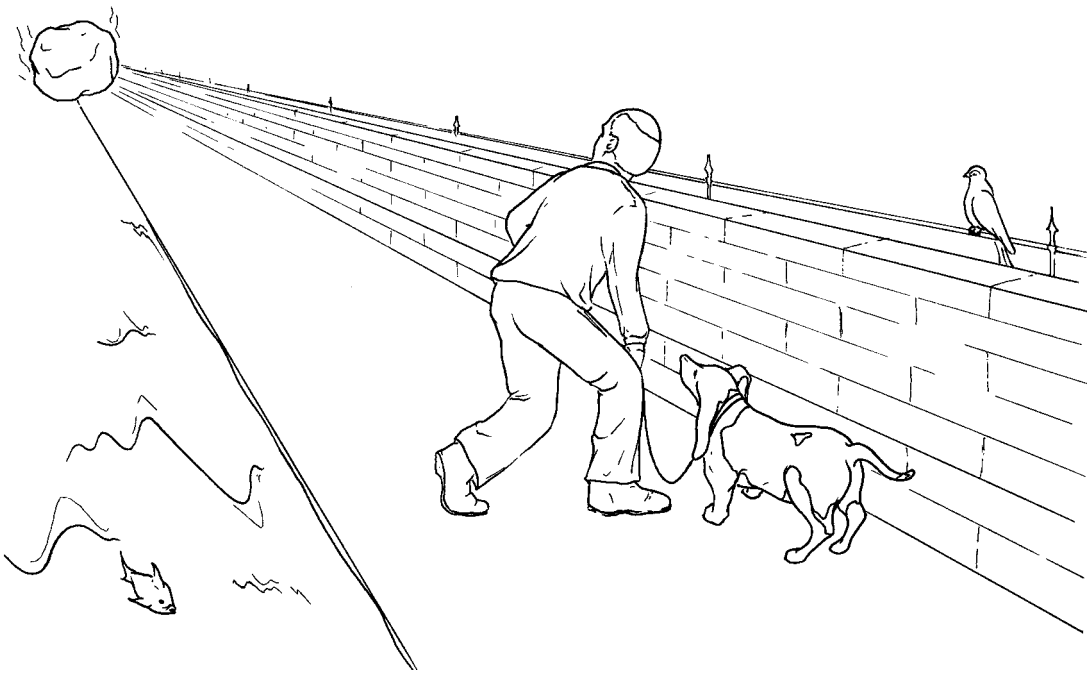
3 This table shows the speed of sound in different materials. Use it to answer the questions.

Material	Speed of sound (m/s)
air	330
water	1500
brick	3000
iron	5000

a In which of the materials does sound travel fastest?

b Does sound travel faster in air or in water?

c In which material does sound travel twice as fast as in water?



d Look at the picture. Order these statements, from the one who is first to hear or feel the sound vibrations to the one who is last. Write the letters in order.

A The person on the path hears the sound.
B The dog hears the sound through the brick wall.
C The bird on the railing feels the vibration through its feet.
D The fish feels and hears the sound through the water.

4 The Sun is a huge, hot, bright ball of gas. We can see it easily, but we can't hear it. A student says that is because it is a long way away, but we can still see it.

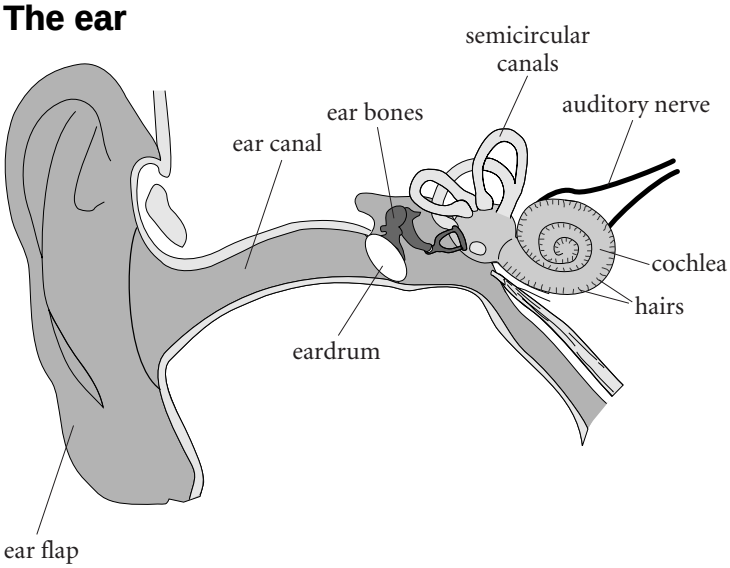
a Why can't we hear the Sun burning?

b Science fiction films are not always very good at showing how sound behaves. Give some examples of things that make a lot of noise on Earth but would happen silently in space.

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How do we hear sounds?

The ear



1 Copy the flow diagram at the side of the page. Use the diagram of the ear to help you fill in the gaps.

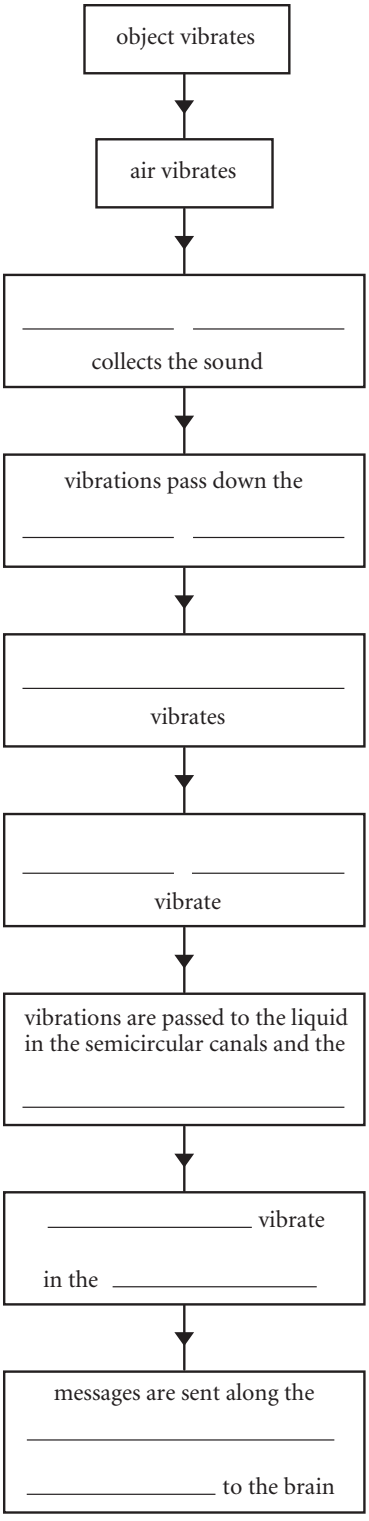
2 Read this description of animals using ultrasound, and answer the questions.

Ultrasound

Ultrasound is the name we give to sound that is too high pitched for humans to hear. The frequency or number of vibrations each second is so high that our ears will not detect the vibrations. The first instruments for detecting ultrasound were developed in about 1938. They could detect frequencies up to 75 000 Hz (vibrations per second). Most humans can only detect frequencies up to 15 000 or 20 000 Hz.

These instruments detected signals from bats and from insects, which no one knew existed before. There are many singing insects we never hear, and more are being discovered.

To make a sound, an animal has to produce a vibration. Very small animals can only make high-pitched sounds because they are not big enough to produce low-frequency vibrations. Some people can hear the high-pitched squeak of a mouse, but rats, mice, voles and hamsters make lots of much higher-pitched sounds when they are communicating with each other. One advantage of this is that some of their predators cannot hear such high frequencies, although cats can hear to 90 000 Hz.



a What is the highest frequency humans can hear?

b Can cats hear higher frequencies than humans?

c What is ultrasound?

d When was ultrasound first discovered?

e A short string gives a higher note than a long string. How does this help to explain why insects make high-pitched sounds?

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Can sound be dangerous?

Our ears can hear a very wide range of sounds, from very quiet to very loud. But above a certain level the ear is damaged by the sound.

This table gives a guide to the loudness of different sounds measured in decibels.

Typical sound level (decibels)	Sound
0	silence
30	ticking watch
30	whisper
60	normal conversation 1 m away
70	busy street
80	alarm clock
80	inside a small car
90	automatic lathe
90	heavy truck
100	food processor at 1 m
110	car horn at 6 m
110	power mower at 1.5 m
110	pop group at 1.5 m
120	jet engine at 150 m
130	pneumatic riveter or hydraulic press

1 **a** What is the sound level given for a car horn at 6 m?

b Explain why some of the sounds have distances given, such as ‘at 6 m’.

c Why do you think the table says ‘typical sound level’ rather than just ‘sound level’?

2 **a** Choose ten sounds from the table and plot them on a bar chart to show the different sound levels. Make sure the scale goes up to at least 120 decibels.

b Sound levels above 80 decibels are dangerous. The more time you spend hearing sounds above this level, the more likely your ears are to be damaged in the long term.

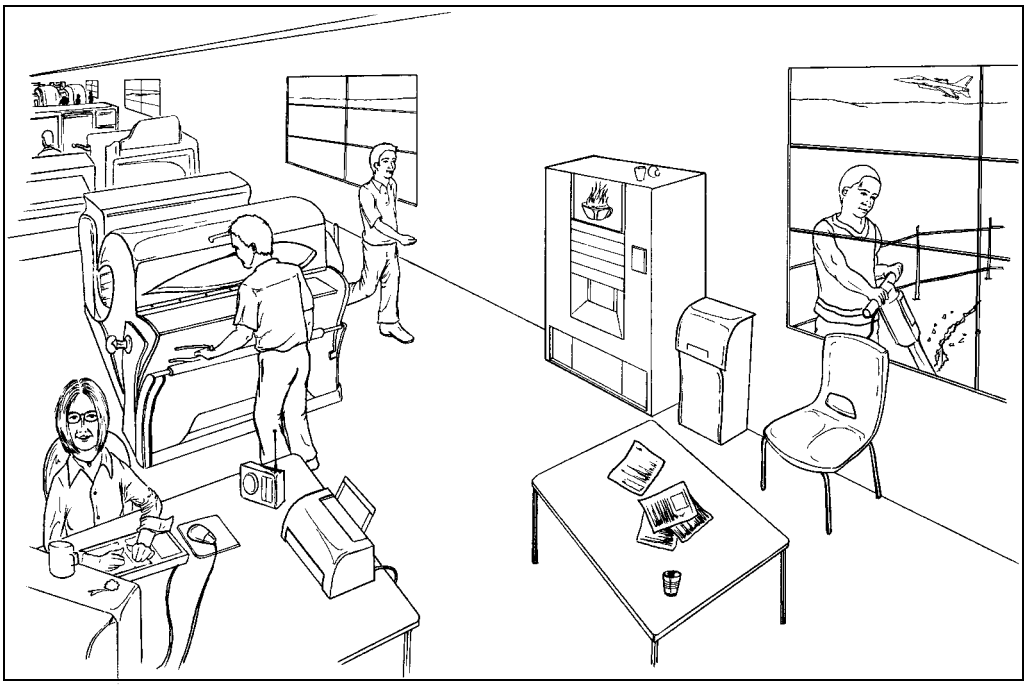
Draw a red line marking the 80-decibel line across your bar chart, and label it ‘dangerous level’.

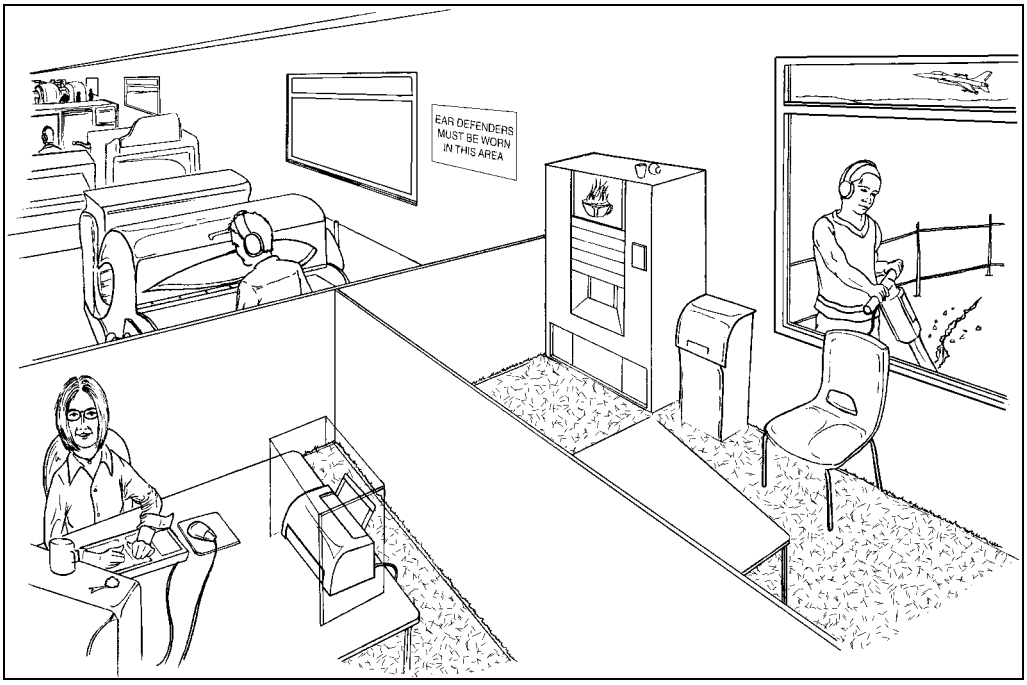
c Which sounds on your bar chart are above this dangerous level?

d Sound levels above 120 decibels are painful to the ears. Draw a line at 120 decibels across your bar chart, and label it ‘painful level’.

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Ear protection





The first picture shows a very noisy working area. In the second picture, changes have been made to reduce the noise levels.

Write down all the changes you can see and explain how each one has helped reduce danger to people's ears.

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